

Digital Logic Design

Spring 2025

Assignment No. 4 (CLO-2,3) Semester Project

Class: BCE-2

Total Marks: 10 Marks

Smart Parking Lot Management System

Produce and simulate a Smart Parking Lot Controller using digital logic circuits. The system should monitor available spaces, control gate access, and display occupancy using digital outputs.

Core Features to Implement:

1. Entry & Exit Control

- Use sensors (simulated with switches) to detect a car at the **entry** or **exit** gate.
- Only allow entry if space is available (based on internal counter).
- Entry/exit gate signal (e.g., LED or output pin) should activate appropriately.

2. Space Counter

- A **binary counter** tracks how many parking spots are currently occupied.
- If full, block entry and turn on a **"FULL"** indicator (LED).
- If empty, block exit to avoid negative count.

3. Occupancy Display

- Show current occupancy using:
 - **4-bit binary output**
 - **7-segment display** for human-readable count

4. Manual Override & Reset

- Include an **admin switch** to reset the system or override it (e.g., for emergencies).

Deliverables

- Working Hardware
- Proteus Simulation
- Verilog Simulation

- Report (System overview, block diagram, logic components, truth tables, state diagrams, k-map simplifications, design summary)

Assessment Rubric

Criteria	Weight
Design Correctness	20
Hardware	30
Simulation and Functionality	30
Report	10
Presentation	10